

query-tuning

Query Tuning Prompt Chain

Use Case

Use this chain when a human analyst needs Edge Copilot to diagnose a slow Oracle SQL query, interpret plan evidence, propose safe rewrites, and review timing or equivalence results.

Required Local Preparation

- Prepare compressed schema context with relevant tables, row counts, keys, indexes, predicates, and join columns.
- Have the original slow query and observed runtime context ready.
- Capture **EXPLAIN PLAN** output before asking for rewrites.
- Keep representative bind values, row counts, and timing notes available for verification.

Prompt Sequence

1. 01-intake.md: establish task scope and ask Copilot what context it needs.
2. 02-query-context.md: submit compressed schema context and the slow query.
3. 03-explain-plan.md: submit plan output and ask for evidence-based bottleneck analysis.
4. 04-rewrite-request.md: ask for semantically safe rewrite candidates.
5. 05-verification-review.md: submit timing and equivalence results for review.

Related Procedure Closet

- `procedures/query_tuner/SKILL.md`

Related Reference Drawers

- `references/oracle-sql/sections/query_tuning.md`
- `references/oracle-sql/sections/oracle_idioms.md`

Query Tuning Chain - 01 Intake

Follow the MDCS procedure closet at `procedures/query_tuner/SKILL.md`.

Act as an Oracle DBA focused on lowering runtime without changing business results. This is the first step of a multi-turn query tuning workflow.

Before recommending any rewrite, ask me only for the missing inputs needed to diagnose the query safely. Use this intake structure:

1. Tuning objective and current runtime symptoms.
2. Required compressed schema context.
3. Required slow query details.
4. Required diagnostic evidence, including `EXPLAIN PLAN`.
5. Verification data needed later for semantic equivalence and timing review.

Reference `references/oracle-sql/` only when the procedure says it is relevant. Do not provide generic tuning advice yet.

Query Tuning Chain - 02 Query Context

Continue following `procedures/query_tuner/SKILL.md`.

I am providing the compressed schema context and the slow Oracle SQL query. Analyze only what can be inferred from this context, then tell me what plan evidence is still needed before rewrite recommendations.

Compressed schema context:

<paste compressed schema context here>

Slow query:

<paste slow Oracle SQL **query** here>

Observed runtime context:

<paste runtime, bind shape, date window, row-volume notes, or SLA here>

Return:

1. A concise restatement of the tuning objective.
2. Relevant tables, joins, predicates, and indexes to watch in the plan.
3. Specific `EXPLAIN PLAN` or execution evidence I should paste next.
4. Any semantic risks that must be preserved during rewrite.

Query Tuning Chain - 03 Explain Plan

Continue following `procedures/query_tuner/SKILL.md`.

I am providing the plan evidence for the current slow query. Use `references/oracle-sql/sections/query_tuning.md` as the relevant drawer, but do not repeat the drawer content back to me.

Plan output:

<paste EXPLAIN PLAN or DBMS_XPLAN output here>

Additional evidence:

<paste estimated vs actual row notes, bind values, waits, timing runs, or known indexes here>

Return:

1. The top bottlenecks, tied directly to plan operations.
2. The likely causes, with confidence level and assumptions.
3. Candidate optimization levers to test.
4. The minimum context still needed before generating rewrites.

Query Tuning Chain - 04 Rewrite Request

Continue following `procedures/query_tuner/SKILL.md`.

Using the compressed schema, slow query, and plan evidence already provided, generate rewrite candidates now.

Return at least two semantically equivalent Oracle SQL alternatives. For each candidate include:

1. The rewritten SQL.
2. Which plan bottleneck it targets.
3. Why it may improve runtime.
4. Semantic risk checks I must run.
5. Expected plan-level signals if it works.

Keep the response concise and evidence-first. Reference `references/oracle-sql/sections/oracle_idioms.md` only if an Oracle idiom is needed for a rewrite.

Query Tuning Chain - 05 Verification Review

Continue following `procedures/query_tuner/SKILL.md`.

I am providing verification results for the original query and rewrite candidates. Review whether the rewrite is safe to recommend.

Verification results:

<paste row counts, aggregate checks, sampled row parity, plan deltas, and repeated timing results here>

Return:

1. Pass/fail assessment for semantic equivalence.

2. Runtime comparison summary.
3. Plan delta summary.
4. Recommended candidate, if any.
5. Remaining risks or local checks before handoff.